connected to one another with a powered hinged joint, and the whole robot is driven by Intel Euclid and powered by a battery pack (providing three minutes of flight time), mounted along the robot's spine.

In the future, DRAGON will be able to wiggle through the air with 12 interlinked modules. It will use its two ends to pick up objects like a two-fingered gripper.

Technique to produce highest performing inverted perovskite solar cells

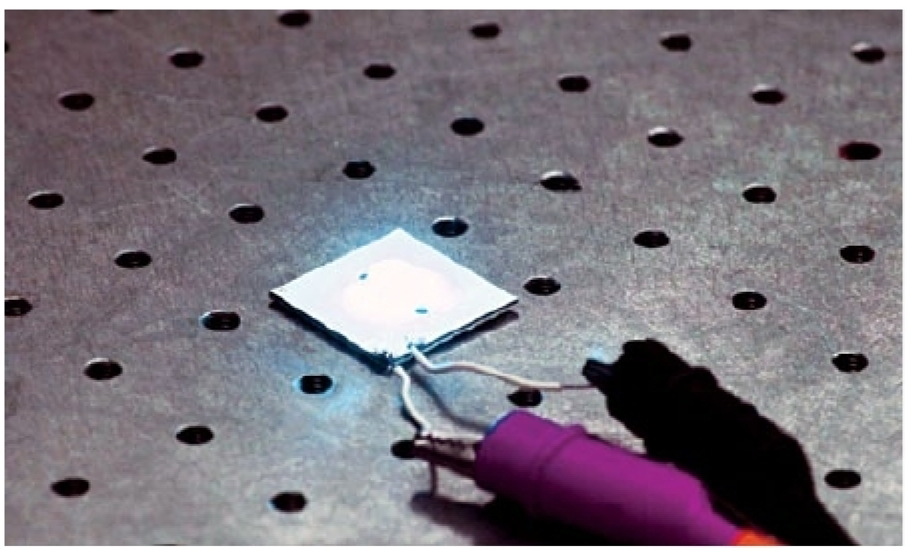
Perovskite-based cells are viewed as next-generation solar cells, offering power conversion efficiency (PCE) performance similar to crystalline silicon-based solar cells. Researchers from Peking University and universities of Surrey, Oxford and Cambridge have come up with a technique that can reduce an unwanted process known as non-radiative recombination, where energy and efficiency is lost in perovskite solar cells.

Researchers created a technique called solution-process secondary growth (SSG), which increases the voltage of inverted perovskite solar cells by 100 millivolts, reaching a high of 1.21 volts without compromising the quality of the solar cells or the electrical current flowing through a device. They tested the technique on a device that recorded a PCE of 20.9 per cent—highest certified PCE for inverted perovskite solar cells till date.

Quantum dot white LEDs achieve record efficiency

Researchers at Koç University, Turkey have demonstrated nanomaterial-based white LEDs that exhibit a record luminous efficiency of 105 lumens per watt. Luminous efficiency is a measure of how well a light source uses power to generate light. With further development, the new LEDs could reach efficiencies over 200 lumens per watt, making these a promising energy-efficient lighting source for homes and offices.

Research leader Sedat Nizamoglu, Koç University, Turkey, says, "Efficient LEDs have strong potential for saving energy and protecting the environment. Replacing conventional lighting sources with LEDs with an efficiency of 200 lumens per watt would decrease the global electricity consumed for lighting by more than half. That reduction is equal to the electricity created by 230 typical 500-megawatt coal plants. It would also reduce greenhouse gas emissions by 200 million tons."



Quantum dot white LEDs (Credit: Optical Society of America)

The LEDs use commercially available blue LEDs combined with flexible lenses filled with a solution of nano-sized semiconductor particles called quantum dots. Light from the blue LEDs cause the quantum dots to emit green and red, which combines with the blue emission to create white light.

The Netherlands to unveil world's first 3D-printed housing complex

The Dutch city of Eindhoven plans to unveil the world's first 3D-printed housing complex in 2019, which could revolutionise the building industry by speeding up and customising construction. Printed in concrete by a robotic arm, the project—named Project Milestone—is backed by the city council, Eindhoven Technical University and several construction companies.

According to Rudy van Gurp, project manager for Project Milestone, a housing complex of five homes of various shapes and sizes will be built over the next three to five years, financed by private investors. He told *Agence France-Presse (AFP)*, "One of the great advantages of 3D printing is that pod-like homes can be completely customised, and even built around natural objects."

Apart from speeding up the building process—from months to weeks—3D printing also solves the problem of scarcity of skilled artisans in The Netherlands, which drives up prices. "In a few years we will not have enough craftsmen like masons. By introducing robotisation to the construction industry, we can make homes more affordable in the future," said van Gurp.

Yasin Torunoglu, deputy mayor, Eindhoven, told *AFP*, "Using 3D printing people have more influence on the design of their house and the way it is built. It is a game-changing innovation."

VR could hold the key for spotting child abuse

A research project carried out over three years, led by a University of Birmingham academic, working with colleagues from Goldsmiths, University of London and University College London, has indicated that virtual reality (VR) could become a vital tool for training general practitioners (GPs) to look out for hard-to-detect signs of child abuse. While some skills can be taught in a straightforward manner, others are harder to impart without the benefit of experience and role models. The ability to pick up signs that a child may be suffering from abuse at home is one such skill that cannot easily be taught.